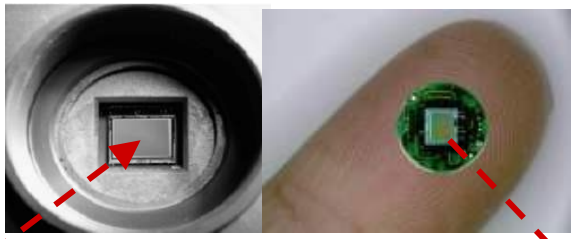
# Machine Vision Basic

COGNEX SUPERHEROES 2013

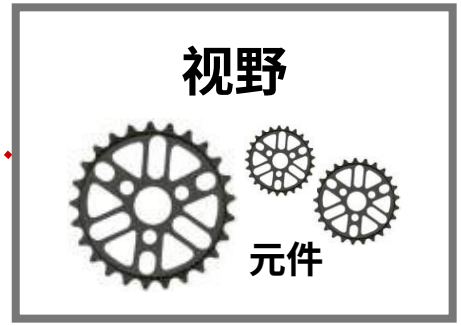
COGNEX

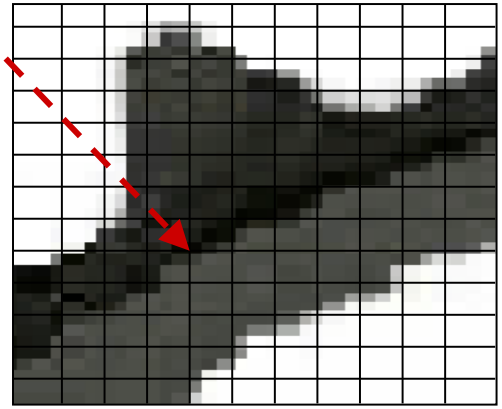
# 成像过程



光信号采集

硬件



结果图像

# 硬件相关术语

*视野（FOV)*

相机所能看到的现实世界的物理尺寸

*工作距离（WD ）*

被测物到相机镜头的距离

*焦距（Focus ）*

透镜中心到其焦点的距离

*感光器尺寸（CCD/CMOS Size ）*

感光片成像区域的尺寸（通常由感光片长边决定）

*景深（DOF ）*

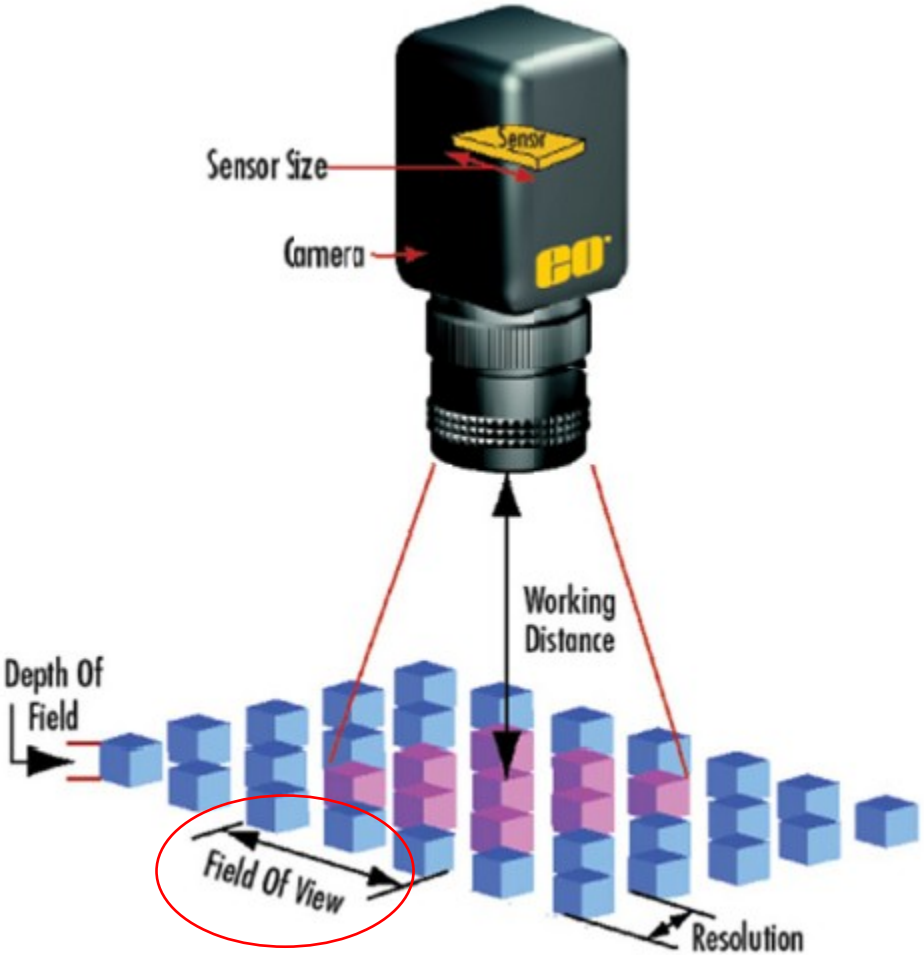
被测物体清晰成像的最上表面与最下表面之间的距离

*放大倍率（MPAG ）*

感光器尺寸与 FOV 之间的比值

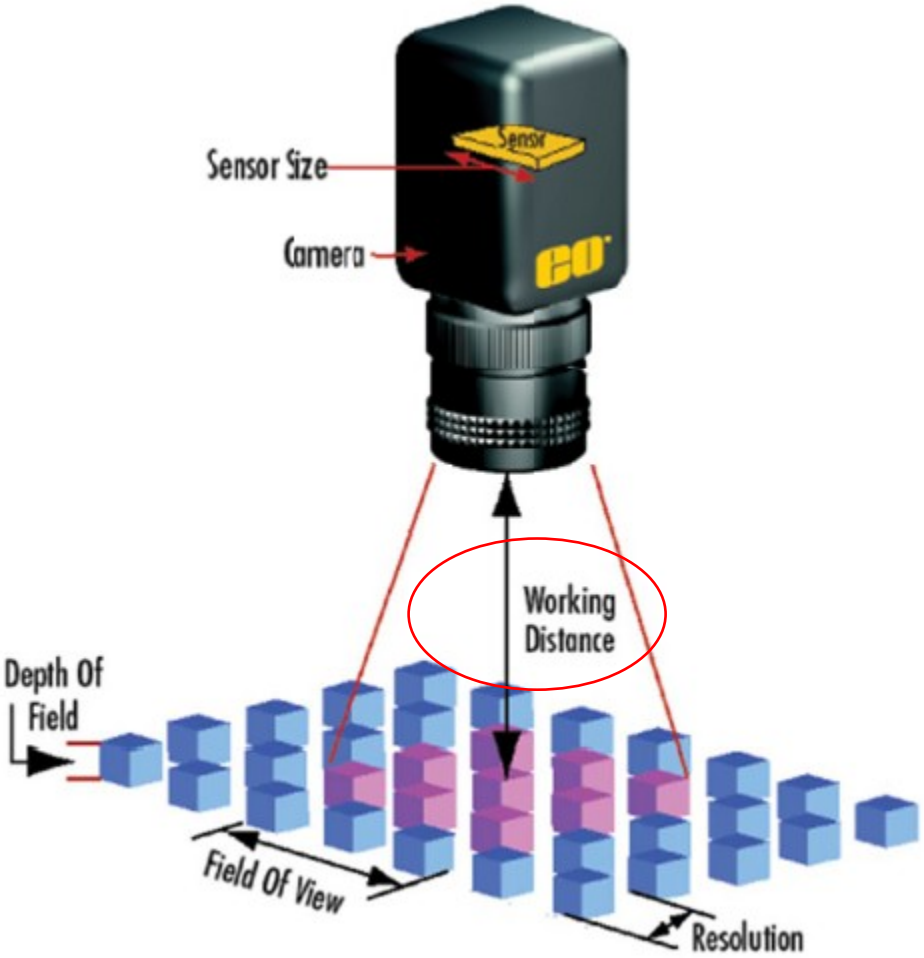
# 视野（FOV)

相机所能看到的现实世界的物理尺寸



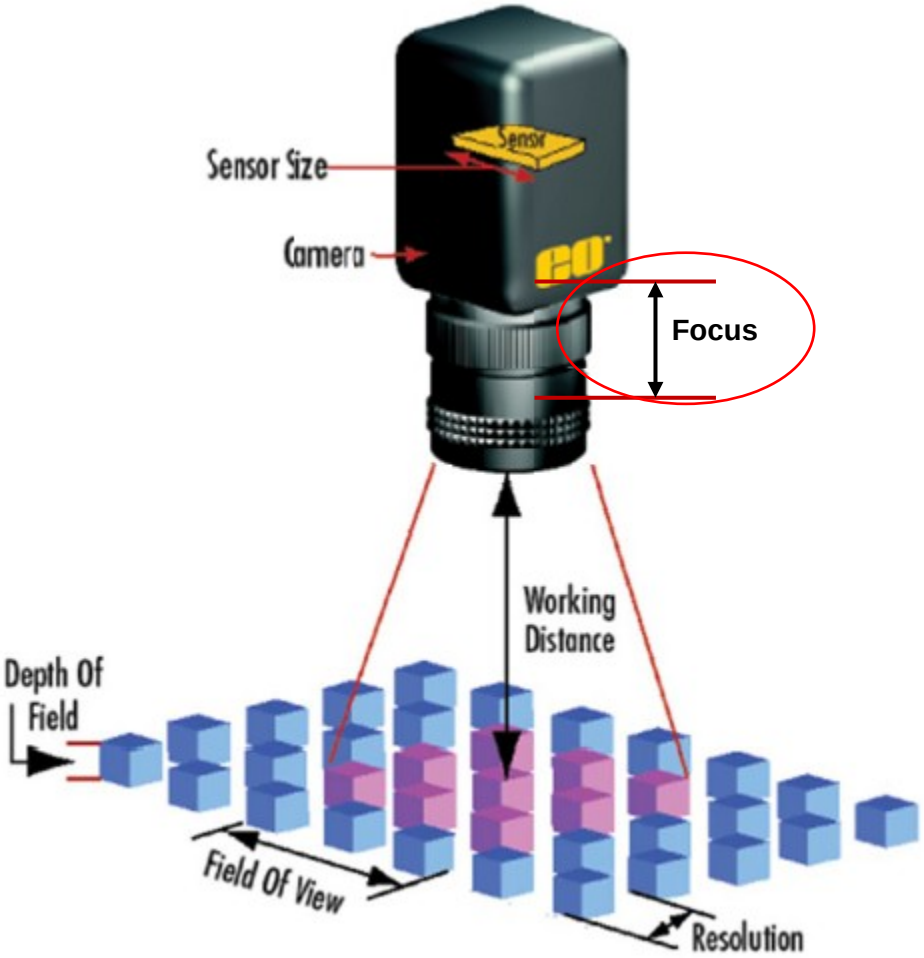
# 工作距离（WD）

被测物到相机镜头的距离



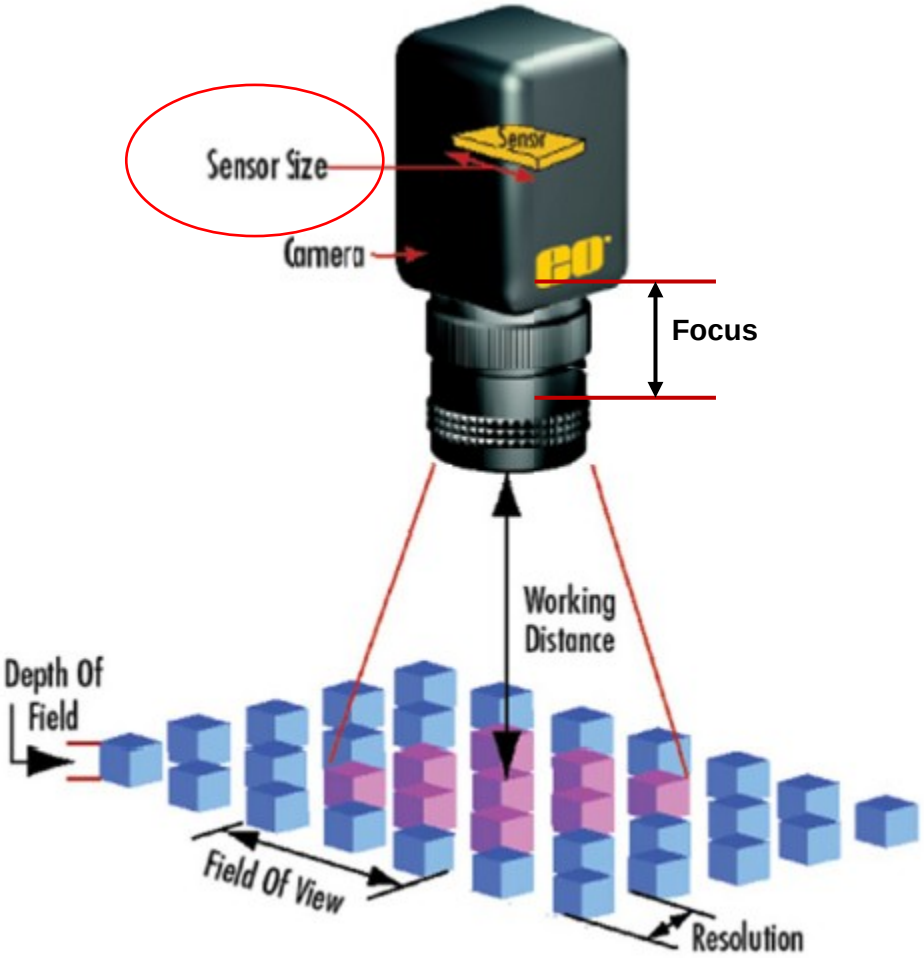
# 焦距（Focus)

透镜中心到其焦点的距离



# 感光器尺寸（Sensor Size）

感光片成像区域的尺寸（通常由感光片长边决定）

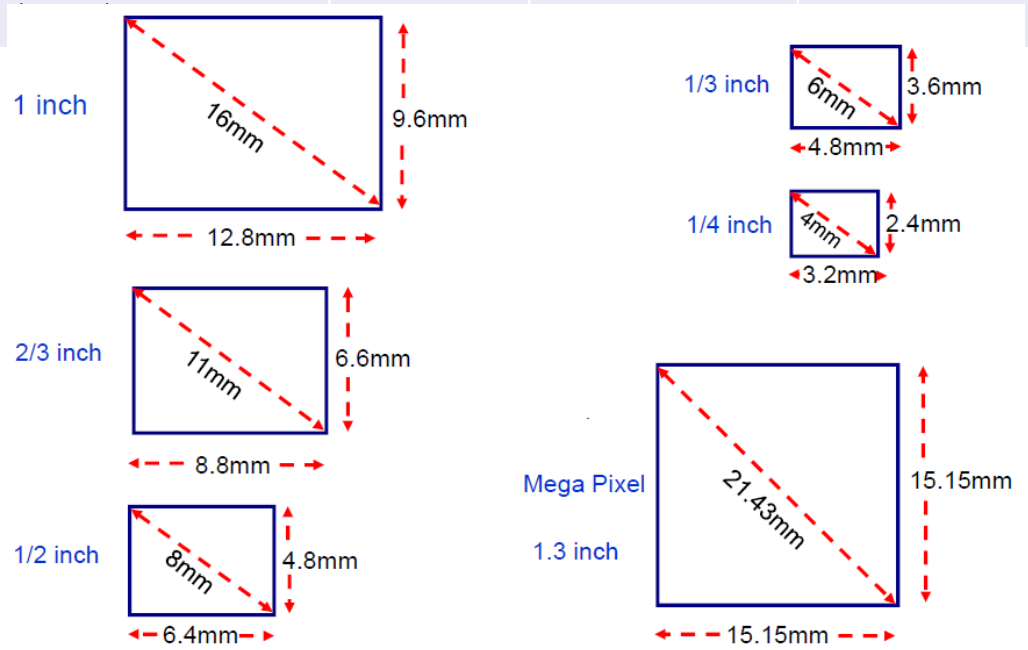


# 感光器尺寸（Sensor Size）

像元尺寸 (Pixel Size)

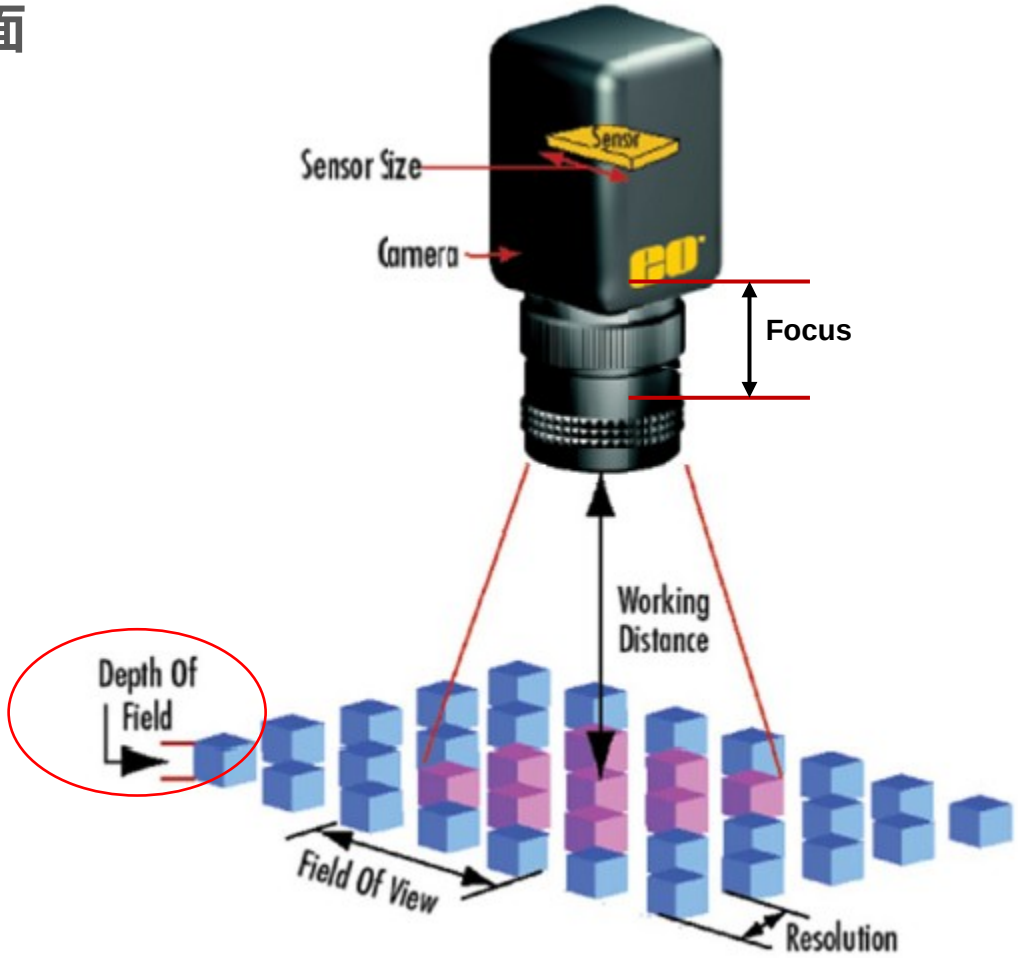
一般为 3μm-10μm 。 (1/3’:7.4X7.4μm　1/1.8’:4.4X4.4μm)

| | 1/4 inch | 1/3 inch | 1/2 inch | 2/3 inch | 1 inch |
|---|---|---|---|---|---|
| 对角线（ mm) | 4 | 6 | 8 | 11 | 16 |
| 幅面尺寸 | 3.6X2.7 | 4.8X3.6 | 6.4X4.8 | 8.8X6.6 | 12.8X9.6 |



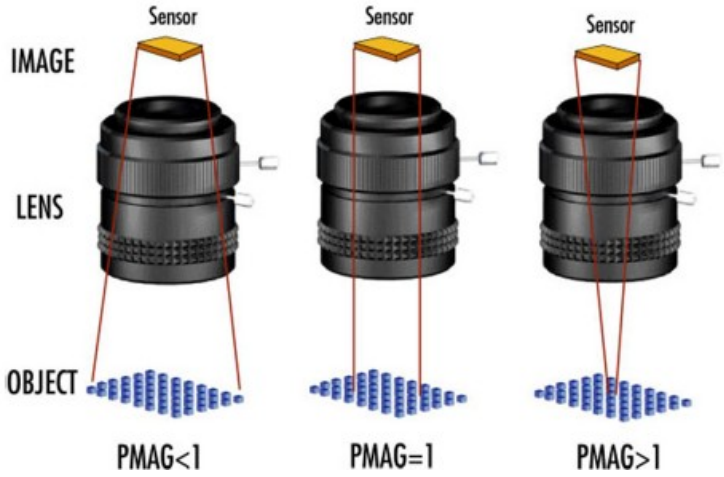
# 景深（DOF）

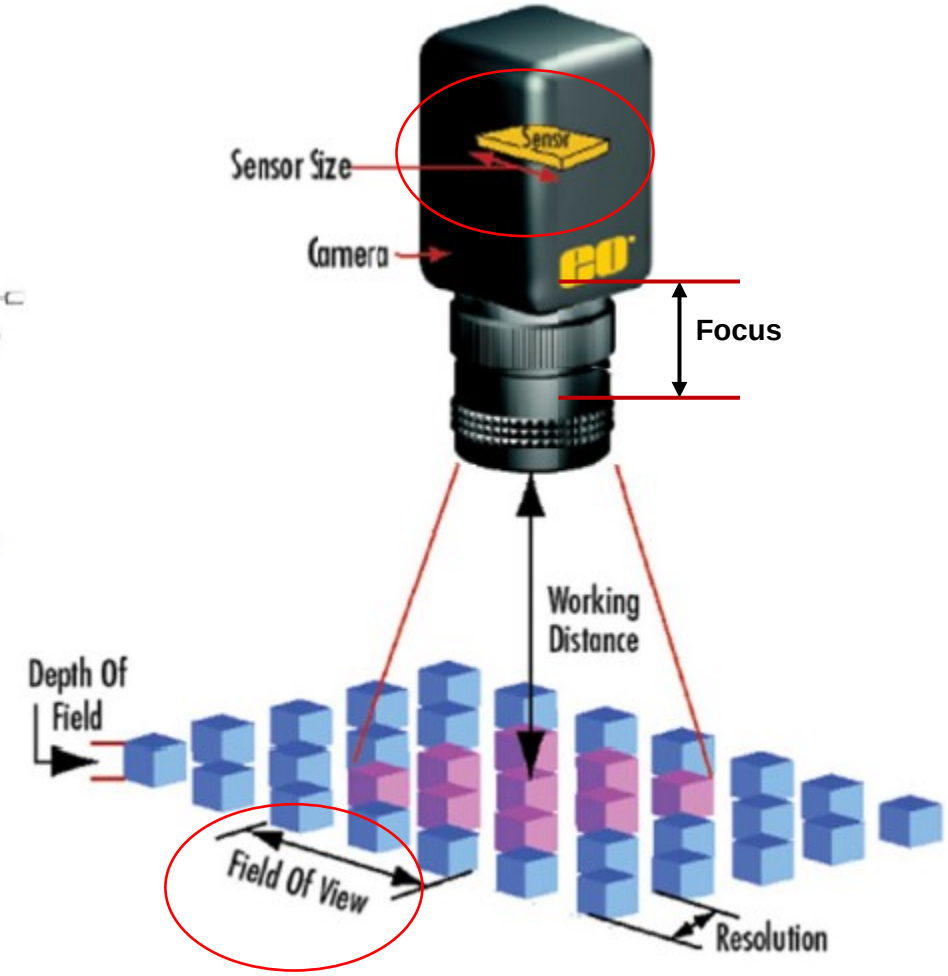
被测物体清晰成像的最上表面
与最下表面之间的距离



# 放大倍率（MPAG）

## 感光器尺寸与 FOV 之间的比值



# 结果图像相关术语

- *像素（Pixel)*
  - 感光器件上的基本感光单元，既相机识别到的图像上的最小单元
- *像素深度(Pixel Depth)*
  - 即每个像素数据的位数，一般常用的是 8Bit ，对于数字工业相机机一般还会有 10Bit 、 12Bit 等
- *分辨率（Resolution)*
  - 相机采集图像的像素点数
- *像素分辨率（mm/pixel)*
  - 每个像素代表的毫米值

# 相机的世界

Can you calculate what this is?

Now maybe you can <u>see</u> it…

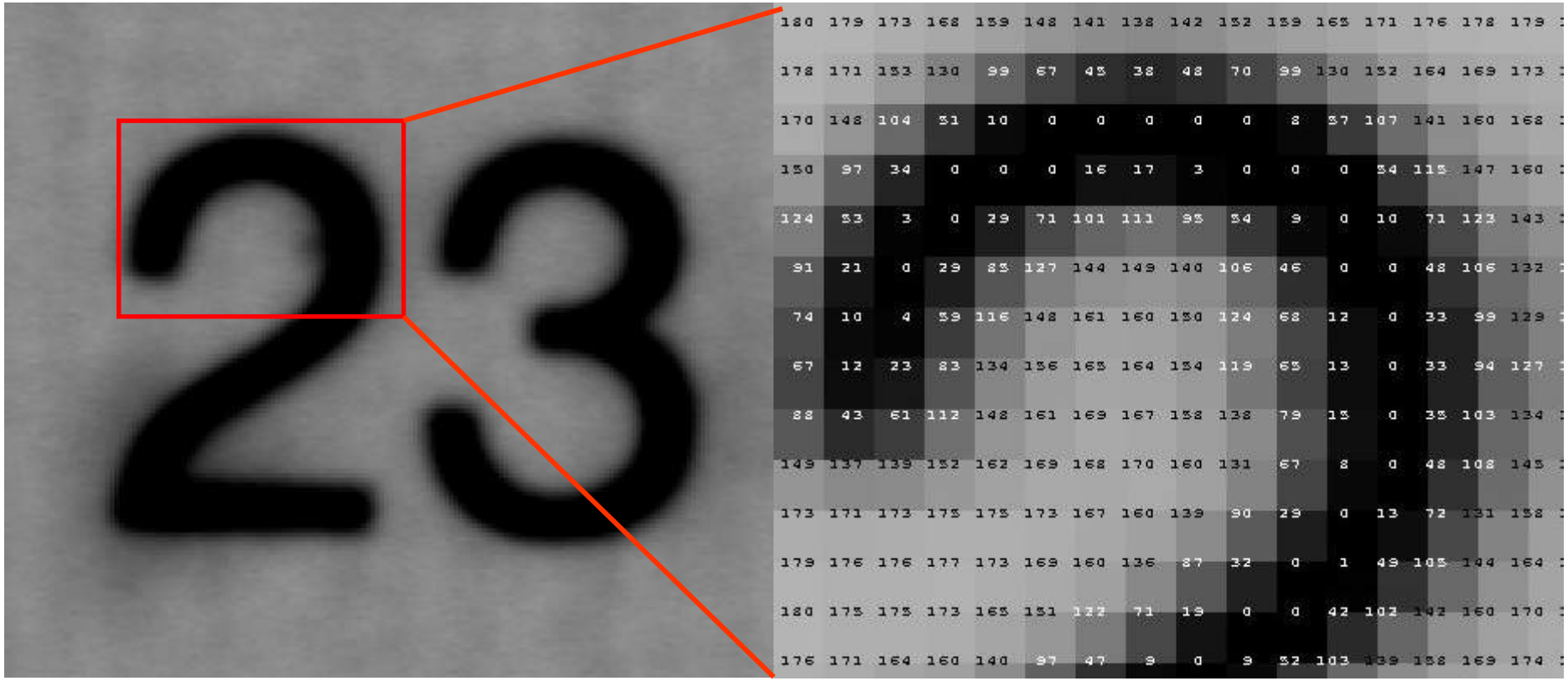
Here it is:

2D Code

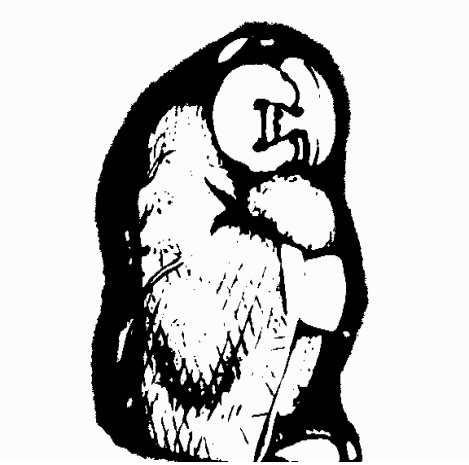
| | | |

# 像素（Pixel)

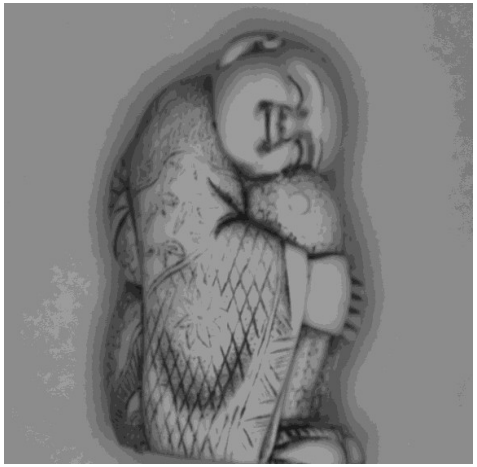
- 一组矩阵亮度值



# 像素深度 (Pixel Depth)



2



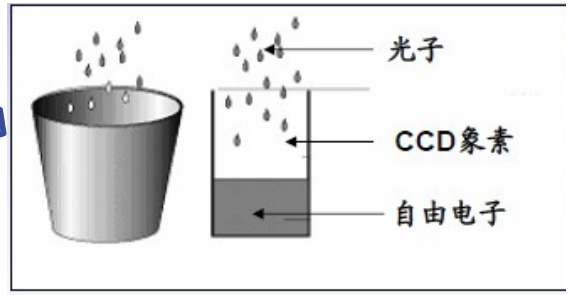
10

128

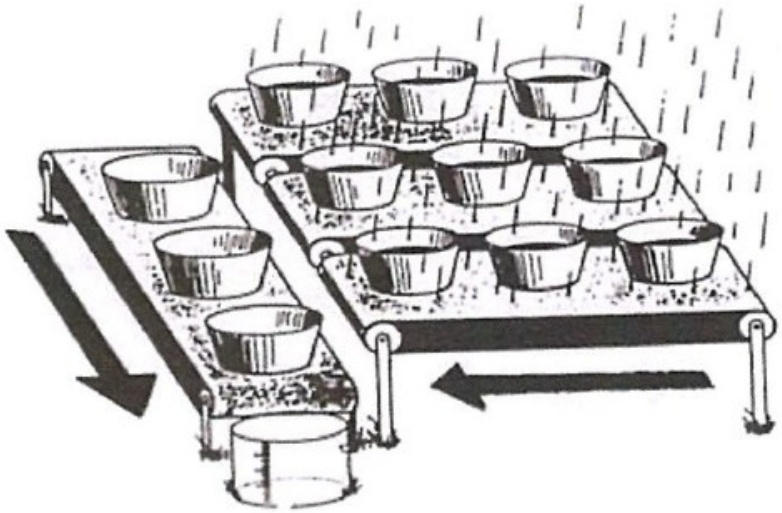
63 = white

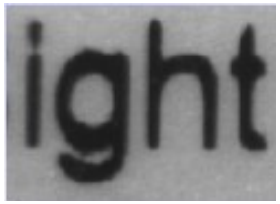
255 = white

# 灰度值 (Gray Value)




| 1 | 2 | 3 | 4 | . | . | . | . | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 2 | | | | | | | | | |
| 3 | | | | | | | | | |
| 4 | | | | | | | | | |
| . | | | | | | | | | |
| . | | | | | | | | | |
| . | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |

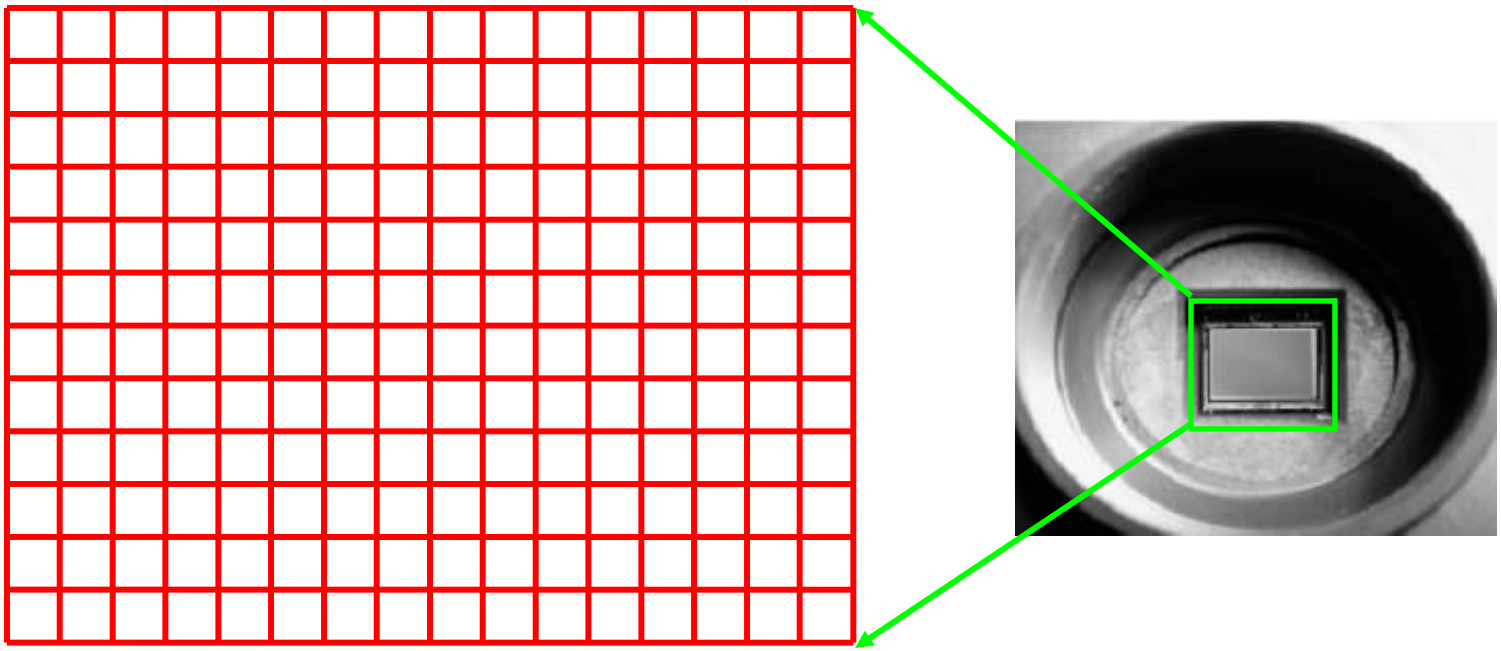




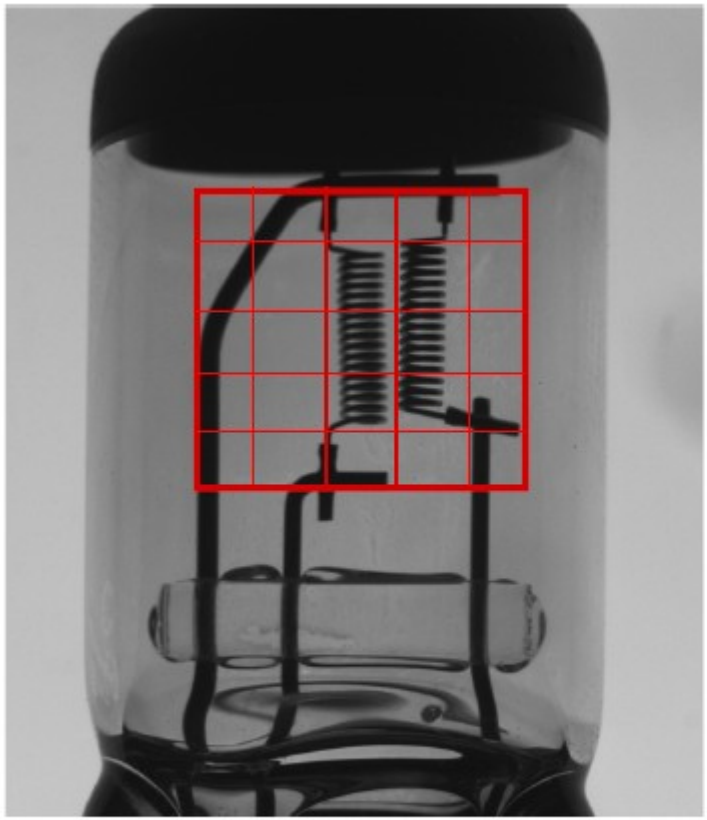

溢出

# 分辨率（Resolution)

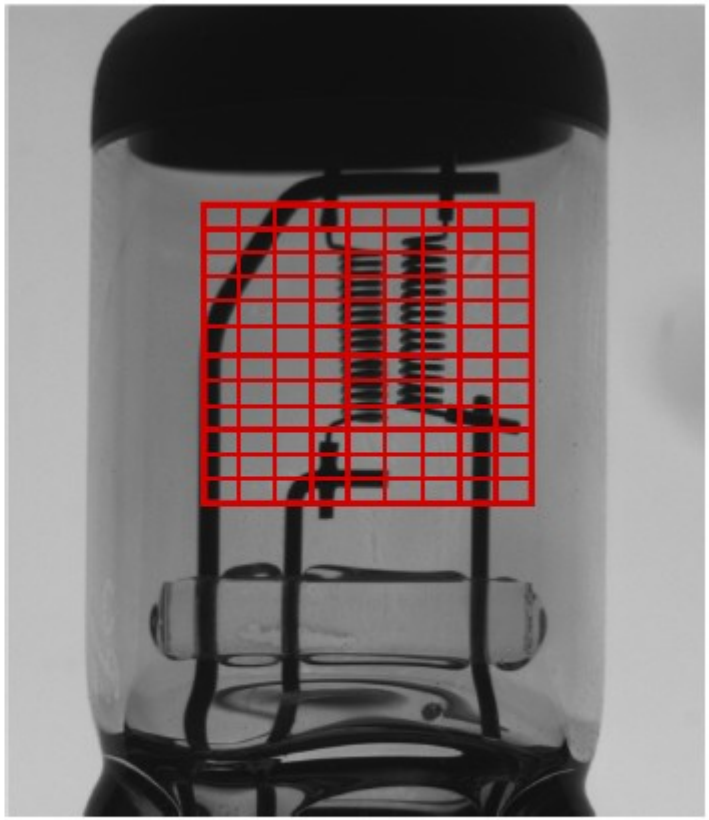
- 图像传感器



# 分辨率（Resolution)



640 x 480 pixel
(0.3 megapixels)

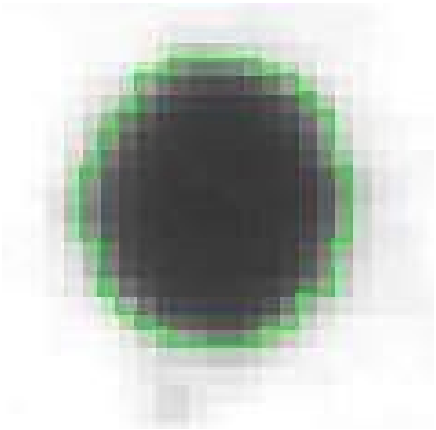


1600 x 1200 pixel
(2 megapixels)

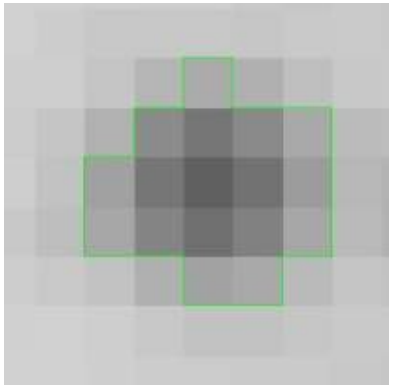
# 分辨率（Resolution)

- **图像及微小特征**

当特征很小时通常很难分辨其几何特性。



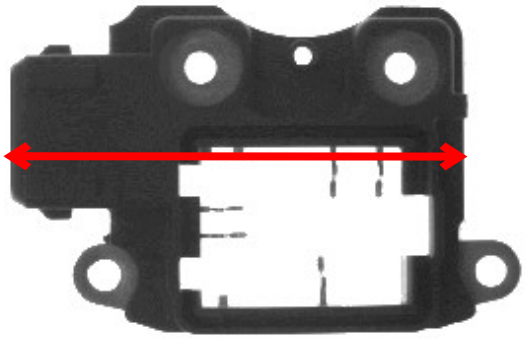
Dot with fine resolution



Dot with coarse resolution

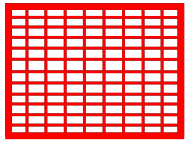
# 像素分辨率（mm/pixel)

- **FOV/Resolution**



**50 x 30 mm Part**

**64 x 48 mmFOV**



**1600 x 1200 pixel**
**(2 Megapixel)**

$$FOV_{horizontal} = 64mm$$

$$Accuracy_{Vision_Tool} = \frac{1}{4}\,pixel$$

$$\#Pixels_{horizontal} = 1600\,pixels$$

$$Accuracy_{horizontal} = \frac{FOV \times Accuracy_{Vision_Tool}}{\#Pixels}$$

$$Accuracy_{horizontal} = \frac{64 \times (\frac{1}{4} \times 2)\,pixel}{1600\,pixels}$$

**Accuracy ≅0.02mm**